

OCR Further Mathematics
Statistics - Chi-squared Tests
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) A practice examination paper is taken by 500 candidates, and the organiser wishes to know what continuous distribution could be used to model the actual time, X minutes, taken by candidates to complete the paper.

The organiser starts by carrying out a goodness-of-fit test for the distribution $N(100, 15^2)$ at the 5% significance level. The grouped data and the results of some of the calculations are shown in the following table.

Time	$0 \leq X < 80$	$80 \leq X < 90$	$90 \leq X < 100$	$100 \leq X < 110$	$X \geq 110$
Observed frequency O	36	95	137	129	103
Expected frequency E	45.606	80.641	123.754	123.754	126.246
$\frac{(O-E)^2}{E}$	2.023	2.557	1.418	0.222	4.280

State suitable hypotheses for the test. [1]

(b) Show how the figures 123.754 and 0.222 in the column for $100 \leq X < 110$ were obtained. [3]

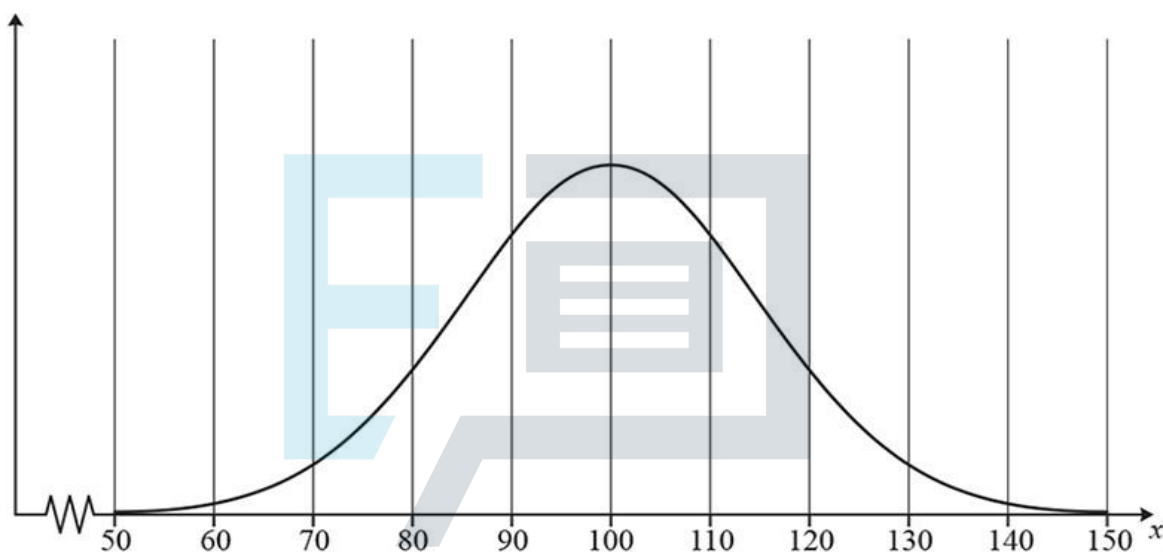
(c) Carry out the test. [4]

(d) The organiser now wants to suggest an improved model for the data.

(i) Suggest an aspect of the data that the organiser should take into account in considering an improved model.

[1]

(ii) The graph of the probability density function for the distribution $N(100, 15^2)$ is shown in the diagram.



On the same diagram sketch the probability density function of an improved model that takes into account the aspect of the data in part (i).

[2]

2(a) A student believes that if you ask people to choose an integer between 1 and 10, not all integers are equally likely to be chosen. The student asks a random sample of 100 people to choose an integer between 1 and 10

inclusive. The observed frequencies O , together with the values of $\frac{(O-E)^2}{E}$ where E is the corresponding expected frequency, are shown in the table.

Integer	1	2	3	4	5	6	7	8	9	10
O	7	8	20	8	7	6	19	7	8	10
$\frac{(O-E)^2}{E}$	0.9	0.4	10.0	0.4	0.9	1.6	8.1	0.9	0.4	0

Show how the value of 8.1 for integer 7 is obtained.

[2]

- (b) Show that there is evidence at the 1% significance level that the student's belief is correct. [5]
- (c) The student wishes to suggest an alternative model for the probabilities associated with each integer. In this model, two of the integers have the same probability p_1 of being chosen and the other eight integers each have probability p_2 of being chosen.

Suggest which two integers should have probability p_1 and suggest a possible value of p_1 . [2]

- 3(a) At a cinema there are three film sessions each Saturday, "early", "middle" and "late". The numbers of the audience, in different age groups, at the three showings on a randomly chosen Saturday are given in Table 1.

Observed frequencies		Session		
		Early	Middle	Late
Age group	< 25	24	20	40
	25 to 60	4	2	10
	> 60	28	22	10

Table 1

The cinema manager carries out a test of whether there is any association between age group and session attended.

Show that it is necessary to combine cells in order to carry out the test. [2]

- (b) It is decided to combine the second and third rows of the table. Some of the expected frequencies for the table with rows combined, and the corresponding contributions to the χ^2 test statistic, are shown in the following incomplete tables.

Expected frequencies		Session		
		Early	Middle	Late
Age group	< 25	29.4	23.1	
	≥ 25	26.6	20.9	

Table 2

Contribution to χ^2		Session		
		Early	Middle	Late
Age group	< 25	0.9918	0.4160	
	≥ 25	1.0962	0.4598	

Table 3

Complete Tables 2 and 3 above.

[3]

(c) Carry out the test at the 5% significance level.

[5]

(d) Use the figures in your completed Table 3 to comment on the numbers of the audience in different age groups.

[2]

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- 4(a) In a standard model from genetic theory, the ratios of types a , b , c and d of a characteristic from a genetic cross are predicted to be 9:3:3:1. Andrei collects 120 specimens from such a cross, and the numbers corresponding to each type of the characteristic are given in the table.

Type	a	b	c	d
Frequency	51	33	30	6

Andrei tests, at the 1% significance level, whether the observed frequencies are consistent with the standard model.

State appropriate hypotheses for the test.

[1]

- (b) Carry out the test.

[6]



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(c) State with a reason which one of the frequencies is least consistent with the standard model. [1]

(d) Suggest a different, improved model by changing exactly two of the ratio values. [1]



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- 5 The table shows the results of a random sample drawn from a population which is thought to have the distribution $U(20)$.

Range	$1 \leq x \leq 8$	$9 \leq x \leq 12$	$13 \leq x \leq 20$
Observed frequency	12	y	$28 - y$

Find the range of values of y for which the data are not consistent with the distribution at the 5% significance level. [9]



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- 6 The numbers of students taking A levels in three subjects at a school were classified by the year in which they entered the school as follows.

	Subject	Mathematics	English	Physics
Year of Entry	Year 7	17	16	7
	Year 12	13	2	5

The Head of the school carries out a significance test at the 10% level to test whether subjects taken are independent of year of entry.

- (a) Show that in carrying out the test it is necessary to combine columns.

[2]



- (b) Suggest a reason why it is more sensible to combine the columns for Mathematics and Physics than the columns for Physics and English.

[1]

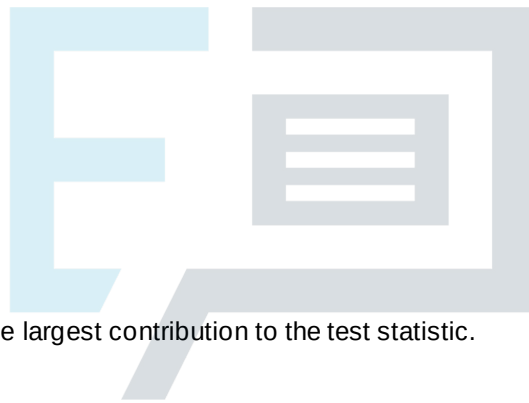
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- (c) Carry out the test.

[6]



(d) State which cell gives the largest contribution to the test statistic.

[1]

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(e) Interpret your answer to part (d).

[1]

- 7 Josh is investigating whether sticking pins into a map at random, while blindfolded, provides a random sample of regions of the map. Josh divides the map into 49 squares of equal size and asks each of 98 friends to stick a pin into the map at random, while blindfolded. He then notes the number of pins in each square. To analyse the results he groups the squares as shown in the diagram.

D	D	D	D	D	D	D
D	C	C	C	C	C	D
D	C	B	B	B	C	D
D	C	B	A	B	C	D
D	C	B	B	B	C	D
D	C	C	C	C	C	D
D	D	D	D	D	D	D

The results are summarised in the table.

Region	A	B	C	D
Number of squares	1	8	16	24
Number of pins	6	21	33	38

- (a) Test at the 10% significance level whether the use of pins in this way provides a random sample of regions of the map.

[10]



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(b) What can be deduced from considering the different contributions to the test statistic?

[1]

- 8 The discrete random variable X is equally likely to take values 0, 1 and 2.
 $3N$ observations of X are obtained, and the observed frequencies corresponding to $X = 0$, $X = 1$ and $X = 2$ are given in the following table.

x	0	1	2
Observed frequency	$N - 1$	$N - 1$	$N + 2$

The test statistic for a chi-squared goodness of fit test for the data is 0.3. Find the value of N .

[4]



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- 9 Carl believes that the proportions of men and women who own black cars are different. He obtained a random sample of people who each owned exactly one car. The results are summarised in the table below.

	Black	Non-black
Men	69	71
Women	30	55

Test at the 5% significance level whether Carl's belief is justified.

[8]



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END OF QUESTION PAPER